

Bloomberg Equity LQA — First Principles Explainer

A section-by-section rebuild of the methodology, with worked numeric examples, so you can reconstruct and reason about it independently.

0. The problem in one sentence

If I want to sell (or buy) V shares of a stock, how much will it cost me (in % of price), how long will it take, and how much uncertainty surrounds both answers — and how do cost, time, and volume trade off against each other?

Everything in the paper is building machinery to answer that one question in a statistically rigorous, calibrated way, for any stock, at any time, for any target volume. The output is not a single number — it’s a **joint probability distribution over (cost, time) given volume**, from which you can extract means, percentiles, and uncertainty bands.

1. Building blocks (Section 1 of the paper)

Before any modeling, you need vocabulary. These are not modeling choices — they’re just definitions:

Term	Definition	Why it matters
Mid-Market Price (S)	Average of best bid and best ask	The reference point from which all “cost” is measured
Market Price Risk (σ)	Daily volatility of the mid price — the stock moves even if you do nothing	This is <i>noise</i> you must separate from <i>impact you cause</i>
Trade Volume (V)	Shares you want to liquidate	The thing you control
Expected Daily Volume (V_E , EDV)	20-day average daily volume	The market’s natural capacity — your denominator for “how big is my order relative to the market”
Participation Rate (π)	Your volume as a % of market volume during execution	$\pi = V/V_T$ — how aggressively you’re trading
Meta-order	An institutional order broken into many smaller trades	Real-world orders aren’t one trade — they’re a <i>sequence</i> , and that sequence has a footprint

Term	Definition	Why it matters
Market Impact	The price change attributable to <i>your</i> order specifically, net of everything else moving the market at the same time	The central quantity to be modeled

First-principles intuition: if you dump a large order into the market all at once, you consume the available liquidity (bids stacked at the best price) and push the price against yourself. If you spread the same volume out over a longer horizon, the market can “refill” between your trades, so a given V costs less the longer T you allow. That single trade-off — **cost falls as time rises, for fixed volume** — is the whole reason a joint (cost, time, volume) model exists instead of a single “cost per share” number.

2. The Transient Market Impact Model (Section 2) — why it's *transient*

2.1 The empirical puzzle it solves

Two empirical facts are hard to reconcile: 1. The **sign** of trades (buy vs sell) is *positively autocorrelated* — buy orders tend to follow buy orders for a while (institutional orders get broken into many same-direction child trades). 2. Yet stock **prices are diffusive** (roughly a random walk) — they don't systematically run away in one direction just because recent order flow was one-sided.

If market impact were **permanent** (every trade pushes the price and it stays pushed forever), fact #1 (persistent same-direction flow) would force prices to super-diffuse — trend predictably — which isn't observed and would in fact allow costless price manipulation (you could buy small amounts and profit as everyone else's future buying pushed the price up further).

Resolution: impact must be *transient* — a trade moves the price, but that impact *decays* over time. This is the entire justification for the “decay kernel” $G(\cdot)$ that appears everywhere below.

2.2 Bouchaud/Lillo-Farmer discrete formulation (Eq. 1)

$$S_N = S_{-\infty} + \lambda \sum_{n=-\infty}^{N-1} \epsilon_n v_n^\delta G(N-n)$$

Read this as: *the price after trade N is the sum of every past trade's impact, each one shrunk by how long ago it happened.* $\epsilon_n = \pm 1$ is direction, v_n^δ is a **concave** (sub-linear, since $\delta < 1$ typically) function of trade size — doubling your trade size less than doubles the impact, a well-documented “square-root-like” impact law — and $G(N-n)$ is the decay: older trades (n far below N) contribute less.

2.3 Gatheral's continuous-time version (Eq. 2)

$$S_t = S_0 + \int_0^t f(x_s)G(t-s) ds + \int_0^t \sigma dW_s$$

Same idea, continuous time: price = starting price + (accumulated, decayed impact of your trading rate $f(x_s)$) + (pure random-walk noise σdW_s from everything else in the market). $f(\cdot)$ is the **instantaneous impact function** (how hard *this instant's* trading rate pushes price); $G(\cdot)$ is still the decay kernel.

2.4 The no-arbitrage constraint

Gatheral shows that if you could round-trip (buy then sell) and *expect to profit* purely from your own impact pattern, that's a form of price manipulation — an arbitrage. Ruling this out mathematically forces a relationship between the impact-decay exponent γ (if $G(\tau) = \tau^{-\gamma}$) and the impact-size exponent δ (if $f(v) \propto v^\delta$): $\gamma + \delta \geq 1$. This single inequality is why the paper can later just plug in power-law forms for f and G with confidence they're economically sound, not arbitrary curve-fits.

2.5 Worked mini-example: impact decay in isolation

Suppose $G(\tau) = \tau^{-0.5}$ (so $\gamma = 0.5$) and a single trade injects impact of size 1 unit at $\tau = 0$. The remaining impact at various horizons:

τ (time since trade)	$G(\tau) = \tau^{-0.5}$
0.25	2.00
1	1.00
4	0.50
16	0.25

Impact halves every 4× increase in elapsed time — it never hits exactly zero (a “long memory” / power-law decay, not exponential), but it becomes economically negligible. This is why splitting a large order over a longer horizon reduces realized cost: by the time you place your *last* child order, the impact from your *first* child order has mostly faded.

3. Price Formation in “Volume Time” (Section 2.2)

Key trick: instead of measuring the horizon of your execution in *clock time* (9:31am to 9:47am), the model re-parametrizes time as **cumulative expected market volume traded** — “volume time.” Why? Because market activity is not uniform through the day (it's heavy at the open/close, thin at lunch);

a 16-minute window at 9:31am and a 16-minute window at 12:30pm represent very different amounts of “liquidity opportunity.” Volume time normalizes this away.

Definitions: - $V_E(t)$: expected cumulative shares traded market-wide from open to time t . -

$v_E(t) = V_E(t)/V_E$: normalized volume-time (0 at open, 1 at close, given V_E = full-day expected volume). -

$\pi = V/V_T$: your **average participation rate** — your volume V divided by the market’s expected volume V_T during your execution window. - $\tau = V_T/V_E$: **relative execution horizon** — what fraction of a full trading day your execution window represents.

Price process (Eq. 6):

$$\frac{S_v - S_\alpha}{S_\alpha} = \sigma_E \left[\int_{v_\alpha}^v f(\pi_x) G(v - x) dx + \int_{v_\alpha}^v dW_x \right]$$

— relative price change (from your start price S_α) = daily volatility σ_E scaling a (decayed, accumulated impact term + noise term). The functional forms chosen (Eq. 7), consistent with the no-arbitrage constraint:

$$f(\pi_x) = \eta \cdot \text{sgn}(\pi_x) \cdot |\pi_x|^\delta, \quad G(v) = v^{-\gamma}$$

η is an overall impact-magnitude parameter (this is the thing that’s “large” for illiquid stocks); δ, γ govern the concavity of size-impact and the speed of decay respectively.

4. The Liquidation Cost Formula — the heart of the model (Section 3.1)

Plugging the VWAP (uniform participation rate) case into the machinery above and doing the integral (Eq. 8-9) collapses everything down to one clean, calibratable formula for **expected cost as a fraction of trade value**:

$$I(V, \tau) = \sigma_E \cdot \kappa \left(\frac{V}{V_E} \right)^\delta \tau^\beta$$

where $\kappa = \frac{\eta}{(1 - \gamma)(2 - \gamma)}$ and $\beta = 1 - \delta - \gamma$ are just repackaged versions of the three primitive parameters (η, γ, δ).

Read this formula out loud — this is the single most important equation in the whole paper:

Expected liquidation cost = (stock’s volatility) × (a calibrated liquidity constant) × (your order size as a fraction of daily volume, raised to a power < 1) × (your execution horizon as a fraction of a day, raised to a power).

Every piece is economically intuitive: - **More volatile stocks cost more to trade** (σ_E multiplies everything) — volatile stocks have wider effective spreads and more risk premium demanded by liquidity providers. - **Bigger orders (relative to V_E) cost more**, but *sub-linearly* ($\delta < 1$, typically found around 0.4-0.6 empirically in this literature) — doubling your order size less than doubles your cost, because you can spread it out and the market partially refills. - **Longer horizons cost less in impact terms** — $\beta = 1 - \delta - \gamma$; since realistic $\delta + \gamma$ is often close to but can be above or below 1, β can be small or even negative for very slow trades... but note the *total* cost formula (Eq. 8) has a **second term**, a noise term $\propto \sigma_E \sqrt{\tau/3} \cdot \epsilon$ that **grows** with horizon (market-price-risk exposure accumulates the longer you're exposed to the market). So the real trade-off is: fast execution $\rightarrow$ high impact, low market-risk exposure; slow execution $\rightarrow$ low impact, high market-risk exposure. **This tension is precisely why the model needs both a cost dimension and a time dimension** — there is no free lunch of “just execute infinitely slowly.”

Then Eq. 10 generalizes the deterministic impact I into a *stochastic* impact $\mathcal{I} = I \cdot \epsilon_I$ with ϵ_I log-normal (mean 1) — capturing that even at the *same* (V, τ), realized cost varies across days/market conditions. This gives (Eq. 11, filled from the gap-fill pass):

$$\mathbb{E}[\mathcal{C}] = I(V, \tau, M), \quad \text{Var}[\mathcal{C}] = I(V, \tau, M)^2 \cdot \text{Var}[\epsilon_I] + \frac{1}{3} \sigma_E^2 \tau$$

— mean cost is just the impact curve; variance has a piece from impact uncertainty and a piece from pure horizon-driven market risk.

4.1 Worked numeric example — Liquidation Cost

Assume calibrated parameters (illustrative, order-of-magnitude realistic for a developed-market large-cap): $\eta = 0.6$, $\gamma = 0.5$, $\delta = 0.5$ (giving $\kappa = \frac{0.6}{(0.5)(1.5)} = 0.8$, $\beta = 1 - 0.5 - 0.5 = 0$).

Stock: $\sigma_E = 2\%$ daily volatility, $V_E = 5,000,000$ shares/day.

Question: what's the expected cost of liquidating $V = 250,000$ shares (5% of ADV) over $\tau = 0.2$ (a fifth of a trading day, ≈ 78 minutes)?

$$I = 0.02 \times 0.8 \times (250,000/5,000,000)^{0.5} \times 0.2^0 = 0.016 \times \sqrt{0.05} \times 1 = 0.016 \times 0.2236 = 0.00358$$

$\approx$ **0.358% of trade value**. On a \$50 stock, that's about **\$0.18/share**, or **\$44,750 total cost** on the \$12.5M order.

Now if you slow down to $\tau = 1$ (a full day, same $\beta = 0$ so impact term is *unchanged* at 0.358% in this special case where $\beta = 0$) — but your noise/market-risk term $\frac{\tau^{1/2}}{\sqrt{3}} \sigma_E$ grows from $\frac{\sqrt{0.2}}{\sqrt{3}} \times 0.02 = 0.00516$ to $\frac{1}{\sqrt{3}} \times 0.02 = 0.01155$ — **more than double the price-risk exposure** for the same expected impact. This is the cost/time trade-off in action, made concrete.

5. Two-Period (IS) Execution Strategy (Section 3.2)

The VWAP derivation assumes constant participation rate throughout. Real “Implementation Shortfall” style algos often **front-load** — trade a fraction $f > 0.5$ of volume in the first half of the horizon, $1 - f$ in the second half — trading off “get it done before the market moves against you” vs. “don’t impact the price too much early.”

The mechanics (Eq. 12-14) show that **impact from the first-half trades propagates into the second half** (because the decay kernel G hasn’t fully died out by the time you’re trading in period 2) — this is exactly the “transient” impact idea made operational: your own earlier trading makes your later trading more expensive, for a while.

Intuition check: the paper shows that summing the two-period costs $\kappa_1 + \kappa_2$ recovers the *same* single-period formula (Eq. 9) when you plug $f = 0.5$ (symmetric split) back in — i.e., a uniform trading schedule is the special case $f = 0.5$ of the general two-period (and by extension, multi-period) framework. This is a good internal-consistency check: **the model doesn’t care how you slice the schedule, only about the aggregate** (V, τ) — **as long as the schedule itself is “smooth” (no discontinuous dumps).**

6. Bid-Ask Spread (Section 3.3) — the part impact alone misses

The transient-impact model above describes the cost of *walking the order book* as you trade continuously. But there’s also a **static** cost of crossing the spread, even for the smallest possible trade — half the effective bid-ask spread. The paper models this component separately and *adds* it to the impact term:

$$s_e = \frac{0.5 s_q}{1 + \rho s_q} \quad (\text{Eq. 15})$$

where s_q is the *quoted* spread and s_e is the *effective* (realized, often smaller) spread. Note the functional form: at very small s_q , $s_e \approx 0.5 s_q$ (spread cost $\approx$ exactly half the quoted spread, as textbook microstructure theory says); as s_q grows, the denominator $1 + \rho s_q$ grows too, so s_e grows *sub-linearly* — wide-quoted-spread stocks don’t actually cost proportionally as much to trade as their quotes suggest, because quotes on illiquid names are often stale/wide and not fully “real.” ρ is a market-specific calibrated parameter (Fig. 1 shows it empirically declining as s_q rises, consistently across most regions except Japan, which behaves differently).

Combined cost model actually fitted to data (Eq. in Section 4.2): $I(V, \tau, M) + s_e$ — impact *plus* half-spread, fit jointly against realized order costs $C(O)$ via non-linear least squares.

7. Calibration (Section 4) — market cap as a factor

Developed markets (Americas, EMEA, APAC, Japan) show that **market capitalization matters** beyond volatility and volume alone — Eq. 17 adds a factor $(C/\bar{C})^\nu$ (security's market cap relative to the market average) to the impact formula:

$$I(V, \tau, M) = \sigma_E \cdot \kappa \left(\frac{V}{V_E} \right)^\delta \tau^\beta \left(\frac{C}{\bar{C}} \right)^\nu$$

Intuition: two stocks with *identical* volatility, EDV, and order size can still have different impact if one is a mega-cap and one is a smaller cap — because market cap correlates with depth of the order book beyond what daily volume alone captures (more institutional ownership, more resting limit orders, etc.).

Emerging markets (Section 4.4) don't show this effect significantly — likely because in EM, volume/volatility already dominate and cap effects are swamped by other frictions (less institutional participation generally, wider information asymmetries).

Model parameters are fit separately **per region** — 7 calibrated markets (Developed Americas/EMEA/APAC/Japan; Emerging Americas/EMEA/APAC), because $\eta, \gamma, \delta, \rho, \nu$ genuinely differ by market microstructure, using ~8 months of institutional order data from 600+ clients.

8. Liquidation Horizon Distribution (Section 5) — from deterministic cost to a full joint distribution

So far we've built $\mathbb{E}[\text{cost}]$ as a deterministic function of (V, τ) . But the model's real output is a **probability distribution**, so we need a distribution for **horizon** too, not just cost.

Empirical fact (Fig. 5): participation rates $\pi = V/(T \cdot V_E)$ across real institutional orders are well-fit by an **exponential distribution**, $\pi \sim \text{Expon}(\lambda)$ — which is the special case $\text{Gamma}(1, 1/\lambda)$ of the Gamma family. Since $T = \frac{1}{(V_E/V)\pi}$ is an inverse function of an exponential/Gamma variable, T itself follows an **Inverse-Gamma distribution**:

$$T \sim \text{InvGamma}\left(1, \frac{V}{V_E}\lambda\right)$$

Why does this matter? It means: for a *given target volume* V , the time it will actually take to comfortably execute it is *not a single number* — it's a random variable with a known distribution shape (right-skewed, thick right tail — sometimes it takes much longer than typical, rarely does it finish much faster). Figure 6 shows exactly this: bigger $V/V_E \rightarrow$ the horizon distribution shifts right (longer expected horizon) and gets *more* skewed (fatter tail — bigger orders have more horizon uncertainty, not just higher expected horizon).

Joint distribution (Eq. 19): $p(C, T | V) = p(C | T, V) \cdot \text{InvGammaPDF}(T; 1, (V/V_E)\lambda)$ — cost conditional on horizon and volume, times the horizon’s own marginal distribution. This factorization is the payoff of the whole paper: **you now have a full joint density over (cost, horizon) for any target volume**, from which you can compute: - Expected cost given a target horizon and volume (already had this — Eq. 9/17). - Expected horizon given a target *cost ceiling* and volume — Eq. 20:

$p(T | V, C^*) = p(C^*, T | V) / \int p(C^*, T | V) dT$ — i.e. condition the joint density on the cost slice you’re willing to accept, then look at what horizon distribution results.

This directly produces the primary LQA outputs (Appendix 7.2): given inputs (target cost, target volume) you get output liquidation horizon (and its uncertainty); given (target horizon, target volume) you get output liquidation cost (and its uncertainty); given a target on either, you get a **confidence level** (the percentile at which that target sits in the distribution).

8.1 Worked numeric example — Liquidation Horizon distribution

First, the closed form. For the special case $\alpha = 1$ (which is exactly what the paper derives — $T \sim \text{InvGamma}(1, \beta)$ with $\beta = (V/V_E)\lambda$), the CDF collapses to something you can compute with nothing but `EXP()` :

$$F_T(t) = Q(1, \beta/t) = \exp(-\beta/t) \quad \Rightarrow \quad \text{the } p\text{-th percentile is } t_p = \frac{-\beta}{\ln(p)}$$

(Derivation: since $T = (V/V_E)/\pi$ and $\pi \sim \text{Expon}(\lambda)$, the reciprocal $1/T = (V_E/V)\pi$ is itself exponential with rate $\beta = (V/V_E)\lambda$; a regularized upper incomplete gamma function with shape 1 is just $Q(1, z) = e^{-z}$.)

Calibration check — pick λ so participation rates are realistic. Since $\pi \sim \text{Expon}(\lambda)$ has mean $1/\lambda$, and typical institutional participation rates cluster around 5-10%, a sensible illustrative choice is $\lambda = 12.5$ (mean participation rate $1/12.5 = 8\%$) — **not** a small λ like 0.15, which would imply an (implausible) mean participation rate of 667%.

Take $V_E = 5,000,000$, $V = 250,000$ ($V/V_E = 0.05$), $\lambda = 12.5$: $\beta = 0.05 \times 12.5 = 0.625$.

Percentile p	$t_p = -\beta/\ln(p)$
10th	0.27 days
50th (median)	0.90 days
90th	5.93 days

Now double the order to $V = 500,000$ ($V/V_E = 0.10$): $\beta = 1.25$, and every percentile **exactly doubles** (10th→0.54, median→1.80, 90th→11.87 days) — because t_p is linear in β , which is linear in V/V_E . Note the **ratio** $t_{90}/t_{10} \approx 21.9 \times$ stays *identical* in both cases (this is a scale family in β — doubling the order shifts the whole distribution proportionally right without changing its relative shape). So the precise

statement is: **absolute horizon uncertainty (e.g., the day-gap between 10th and 90th percentile) grows proportionally with order size, even though the relative/multiplicative spread stays constant** — bigger orders take proportionally longer *and* have a proportionally wider absolute uncertainty band, but not a *relatively* more skewed one, in this $\alpha = 1$ special case.

(See the companion Excel workbook, “Equity” tab, for a fully formula-driven version of this Inverse-Gamma calculation, the cost formula, and the confidence-level lookups, so you can flex V , τ , σ_E , V_E and watch every output recompute.)

9. Backtesting (Section 6, Table 1)

The model is validated by comparing **predicted likelihood of observed costs** (using the fitted model) against a naive benchmark (constant impact = the market’s population-average cost). The fitted model beats the naive benchmark by **4-20%** in out-of-sample log-likelihood, with the biggest edge in **Emerging Americas (+20%)** and the smallest in **Developed Japan (+4%)** — i.e., the extra structure in the model (volatility scaling, size scaling, market-cap factor) adds the most value in markets with the most heterogeneity across stocks, and the least value in markets that are already fairly homogeneous/well-approximated by a single constant.

10. Model Inputs & Outputs (Appendix — what you actually see on a Bloomberg terminal)

Inputs (overrides the user sets):

Field	Meaning
LQA_PRICE_VOLATILITY (PY228)	Annualized price vol, from daily log-return std dev
LQA_TGT_LIQUIDATION_VOLUME (PY247)	Your target V
LQA_TGT_LIQUIDATION_COST (PY250)	Your cost ceiling
LQA_TGT_LIQUIDATION_HORIZON (PY253)	Your time ceiling
LQA_EXPECTED_DAILY_VOLUME (PY229)	V_E
LQA_TGT_CONFIDENCE_LEVEL (PY328)	Percentile to evaluate at (e.g. median vs. 90th)
LQA_TGT_PARTICIPATION_RATE (PY326)	Alternative way to specify aggressiveness

Outputs:

Field	Meaning
LQA_LIQUIDATION_COST (PY251)	% cost for given (V, T)
LQA_LIQUIDATION_HORIZON (PY254)	Time to liquidate for given (V, C^*)
LQA_LIQUIDATION_CONFIDENCE (PY246)	Percentile of realizing cost/horizon under target
LQA_LIQUIDATION_COST_UNC (PY252)	Std dev / uncertainty band around cost
LQA_LIQUIDATION_HORIZON_UNC (PY255)	Std dev / uncertainty band around horizon
LQA_LIQUIDITY_SCORE (PY275)	1-100 percentile rank vs. universe (100=most liquid)

11. How this is used in practice — liquidity risk modeling

This is where the machinery pays off for a risk manager (not just a trader):

1. **Position-level liquidity risk:** for any position size, get a defensible, market-calibrated (not rule-of-thumb “X days of ADV”) estimate of the *cost* and *time* to exit, with uncertainty bands — critical for fair-value/NAV impact modeling if a large redemption forces liquidation.
2. **Portfolio liquidity stress testing:** aggregate LQA_LIQUIDATION_HORIZON across all positions at a stressed target cost (or stressed volatility input, since σ_E is a direct multiplier — you can shock it) to answer “if redemptions hit and we must liquidate to raise cash, how many days at what cost, under stressed market conditions?” This is directly the kind of exercise regulators (SEC 22e-4 liquidity risk programs, FCA/BoE stress tests) require of funds.
3. **Liquidity scoring for portfolio construction / limits:** LQA_LIQUIDITY_SCORE gives a comparable, cross-sectional percentile ranking — useful for setting concentration limits (“no more than X% of NAV in securities scoring below the 20th percentile”) without needing to re-derive cost curves per name.
4. **Redemption gate / swing pricing calibration:** the cost distribution (not just the mean) lets you price the *tail* cost of a forced, fast liquidation (low τ , e.g. 1-day) versus an orderly one — directly informs swing factor / anti-dilution levy calibration.
5. **Pre-trade / best-execution analysis:** traders use the cost-vs-horizon curve to choose an execution schedule that matches their actual urgency vs. cost tolerance — walking along the curve rather than guessing.

The one-line mental model to carry forward: *cost and time are two sides of the same liquidity coin, linked by a calibrated power-law impact curve plus a volatility-driven noise term, and the model gives you the whole joint distribution — not just a point estimate — so you can reason about percentiles and tail risk, not just averages.*

Where the gap-filling agents flagged genuine uncertainty (for completeness)

- **Page 15** of the 15-page PDF was never reached by the source video (scrolling stopped at page 14 / References) — so any content there (remaining reference entries for Lillo & Farmer (2004) and Zarinelli et al. (2014), which are cited in-text, plus any disclosures) is not captured. This does not affect any formula or example above — it's purely bibliographic.
- Everything else — every numbered equation, every figure caption, and the full 7-row backtesting table — was independently re-confirmed across two rounds of frame sampling.